

Description and choice data for the domain “Surface area”

Description of the choice domain 21, Surface area

The prompt question and the universe of five response options in the choice domain **Surface area** are as follows. The labels *a*, *b*, *c*, *d* and *e* were not displayed during the experiment and are indicated here to allow cross-referencing with data tables and visualizations below and results in the paper.

These countries are ranked, respectively, 2nd through 6th in terms of surface area, including inland bodies of water. In millions of square kilometres, those surface areas are 9.984, 9.573, 9.525, 8.516 and 7.692.

Which one of the following countries do you think has the greatest surface area, including inland bodies of water?

- Canada
- United States of America
- China
- Brazil
- Australia

The following figure is a screenshot from the actual experiment, with one of the 26 possible menus for this domain.

Surface area

Which one of the following countries do you think has the largest surface area, including inland bodies of water?

☐ Brazil

☐ Canada

☐ Australia

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Figure 1: Screenshot for domain Surface area

Menu A	Choice counts					Choice proportions				
	$N_A(a)$	$N_A(b)$	$N_A(c)$	$N_A(d)$	$N_A(e)$	$\hat{P}_A(a)$	$\hat{P}_A(b)$	$\hat{P}_A(c)$	$\hat{P}_A(d)$	$\hat{P}_A(e)$
$\{a, b\}$	37	3	-	-	-	0.925	0.075	-	-	-
$\{a, c\}$	38	-	2	-	-	0.950	-	0.050	-	-
$\{b, c\}$	-	24	16	-	-	-	0.600	0.400	-	-
$\{a, b, c\}$	31	5	4	-	-	0.775	0.125	0.100	-	-
$\{a, d\}$	40	-	-	1	-	0.976	-	-	0.024	-
$\{b, d\}$	-	30	-	11	-	-	0.732	-	0.268	-
$\{a, b, d\}$	31	9	-	0	-	0.775	0.225	-	0.000	-
$\{c, d\}$	-	-	23	17	-	-	-	0.575	0.425	-
$\{a, c, d\}$	34	-	5	1	-	0.850	-	0.125	0.025	-
$\{b, c, d\}$	-	23	13	4	-	-	0.575	0.325	0.100	-
$\{a, b, c, d\}$	30	3	5	2	-	0.750	0.075	0.125	0.050	-
$\{a, e\}$	38	-	-	-	2	0.950	-	-	-	0.050
$\{b, e\}$	-	24	-	-	16	-	0.600	-	-	0.400
$\{a, b, e\}$	35	3	-	-	2	0.875	0.075	-	-	0.050
$\{c, e\}$	-	-	23	-	17	-	-	0.575	-	0.425
$\{a, c, e\}$	34	-	3	-	3	0.850	-	0.075	-	0.075
$\{b, c, e\}$	-	20	12	-	8	-	0.500	0.300	-	0.200
$\{a, b, c, e\}$	27	7	4	-	2	0.675	0.175	0.100	-	0.050
$\{d, e\}$	-	-	-	14	26	-	-	-	0.350	0.650
$\{a, d, e\}$	36	-	-	0	4	0.900	-	-	0.000	0.100
$\{b, d, e\}$	-	21	-	4	15	-	0.525	-	0.100	0.375
$\{a, b, d, e\}$	27	5	-	2	6	0.675	0.125	-	0.050	0.150
$\{c, d, e\}$	-	-	25	3	12	-	-	0.625	0.075	0.300
$\{a, c, d, e\}$	34	-	5	0	1	0.850	-	0.125	0.000	0.025
$\{b, c, d, e\}$	-	12	17	3	8	-	0.300	0.425	0.075	0.200
$\{a, b, c, d, e\}$	26	5	6	0	3	0.650	0.125	0.150	0.000	0.075

Table 1: Observed choice counts and proportions.

Choice data for domain 21, Surface area

Table 1 shows choice counts and choice proportions for this choice domain. For each menu A and each object $x \in \{a, b, c, d, e\}$, $N_A(x)$ is the number of participants who chose object x from menu A and $\hat{P}_A(x)$ is the corresponding proportion of participants who chose x from A . When $x \notin A$, a dash is displayed.

The following figure displays choice proportions for all doubleton and tripleton menus in Barycentric coordinates. See a full description of this graphical representation in the paper. Each panel shows choice proportions for all doubleton and tripleton menus of a different tripleton subset of $\{a, b, c, d, e\}$. The downward-pointed (blue) triangle shows the set of ternary choice proportions that are compatible with regularity and the three binary choice proportions, on the corresponding tripleton. The upward-pointed (red) triangle shows the set of ternary choice proportions compatible with the multiplicative inequality and the three binary choice proportions.

